Practical 6. Foundations of scientific reading

JXH-1104 Neuroscience of Sport Performance

Dr Germano Gallicchio

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Introduction

Navarro et al. (2013) report their results in tables. In this practical, the values in Tables I and III are turned into figures, and the figures are evaluated with the four questions introduced in the earlier session on data and measurement. The analysis is carried out in Excel.

Aim

To present the results of a published study as figures, and to interpret the figures with reference to the study's research question.

Materials

- A computer
- The article: Navarro et al. (2013), available through the Talis reading list
- Excel

TASK 1 · Entering the data

Create a new blank workbook in Excel for the web. Data are entered with one row per condition and one column per variable, with the variable names in the first row.

Table I: outcomes. Enter the number of hits, misses and failures for each of the three conditions (no goalkeeper, goalkeeper, knowledgeable goalkeeper). The result is a table of three rows of conditions and three columns of outcomes.

Table III: errors. In a separate area of the sheet, enter the absolute error and the variable error relative to the target centre, for each of the three conditions.

Check every value against the article before continuing.

TASK 2 · A figure of the outcomes

Create a clustered column chart of the Table I data, with the conditions on the horizontal axis and one column for each outcome.

The figure requires:

- an axis label for each axis, including the unit (e.g., number of kicks);
- a vertical axis beginning at zero;
- a legend identifying the three outcomes.

If the outcomes, rather than the conditions, appear on the horizontal axis, the rows and columns of the chart's data need to be switched.

Questions. Which outcome differs most across conditions? Which does not appear to differ? Compare the figure with the statistical results reported in the article.

TASK 3 · A figure of the errors

Create a clustered column chart of absolute error and variable error relative to the target centre, for the three conditions, with the same requirements as in Task 2 (unit: cm).

Questions. Which of the two errors increases when a goalkeeper is present? With reference to random and systematic error (session on data and measurement), what does this pattern indicate about where the shots were directed?

TASK 4 · Evaluating the figures

Apply the four questions from the session on data and measurement to each figure:

1. Which variable was measured, and in which units?
2. What does each graphical element represent?
3. Which conditions are being compared?
4. Where does each axis begin, and are error bars shown? If not, why not?

Points for discussion

- Table III reports means but no standard deviations. What information is therefore missing from the figure of Task 3?
- Counts of hits depend on the number of kicks taken. Why is a comparison of counts across conditions acceptable in this study? (Consider the number of kicks per condition.)
- Which of the two figures would be more informative for a coach, and why?

Additional exercises

- Complete any section of the structured reading sheet not finished during the lecture.
- Convert the Table I counts to percentages of the kicks in each condition (each condition comprised 200 kicks), and plot the percentages. Does the figure change in appearance or in meaning?
- Table I also reports the mean ball speed and its standard deviation in each condition. On paper, draw a column chart of mean ball speed with error bars showing ±1 SD, using the values in Table I. Mark the speed of 22 m/s that participants were instructed to exceed, and compare it with the means. (The automatic error bars in Excel for the web are calculated by Excel and do not use the standard deviations reported in the article.)

Additional exercises in R (optional)

These exercises are optional and are not assessed. They repeat parts of this practical in R, which runs here in the web browser without installation. The first time a cell is run, R may take up to a minute to load. Select **Run Code** to run a cell; the code can be edited and run again.

The values in R1 and R2 are taken from the article. The values in R3 are **simulated**.

R1 · Table I as a figure

```
outcomes <- matrix(c(127, 37, 36,
                     89,  69, 42,
                     79,  65, 56),
                   nrow = 3,
                   dimnames = list(c("hit", "miss", "failure"),
                                   c("no goalkeeper", "goalkeeper", "knowledgeable")))
outcomes
barplot(outcomes, beside = TRUE, legend.text = TRUE,
        ylab = "Number of kicks", main = "Outcome by condition")
```

R2 · Absolute and variable error

```
errors <- matrix(c(125, 27,
                   149, 23,
                   164, 24),
                 nrow = 2,
                 dimnames = list(c("absolute error", "variable error"),
                                 c("no goalkeeper", "goalkeeper", "knowledgeable")))
barplot(errors, beside = TRUE, legend.text = TRUE,
        ylab = "Error relative to target centre (cm)")
```

Question. Which error changes across conditions, and which remains approximately constant?

R3 · A paired comparison with simulated participants

The article reports only condition means. The values below are simulated for ten fictitious participants, constructed so that their means equal those in Table III (absolute error relative to the target centre, cm). Each position in the two lists is the same participant.

```
no_gk   <- c(118, 131, 104, 142, 127, 109, 135, 121, 138, 125)
know_gk <- c(170, 149, 165, 175, 134, 158, 163, 185, 179, 162)

c(mean_no_gk = mean(no_gk), mean_know_gk = mean(know_gk))
t.test(know_gk, no_gk, paired = TRUE)
```

Questions. Is the difference statistically significant? Which test did Navarro et al. use instead, and why might a different test be required when there are three conditions?

Expected results

- R1: a clustered bar chart in which hits decrease and misses increase when a goalkeeper is present, while failures change little.
- R2: absolute error increases across conditions (125, 149, 164 cm); variable error remains approximately constant (23–27 cm).
- R3: means 125 and 164 cm; $t = 6.75$, $df = 9$, $p < .001$. The authors used analysis of variance (F), because three conditions were compared at once; a t-test compares only two.

If you cannot access the article

Table I (number of kicks): no goalkeeper, 127 hits, 37 misses, 36 failures; goalkeeper, 89, 69, 42; knowledgeable goalkeeper, 79, 65, 56. Table III (cm, relative to the target centre): absolute error 125, 149, 164; variable error 27, 23, 24.